AP BIOLOGY TOPIC 6.1

DNA & RNA Structure, Chromosomes, Plasmids & Classic Experiments

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-1.K | IST-1.L | ANTIPARALLEL 5′→3′ | CHARGAFF'S RULES

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.1 (DNA & RNA STRUCTURE, CHROMOSOMES, PLASMIDS & CLASSIC EXPERIMENTS)

CED Core Vocabulary

Universal Genetic Storage — DNA is the primary hereditary material in all cellular organisms (prokaryotes & eukaryotes); some viruses use RNA genomes (e.g., retroviruses like HIV,...

Eukaryotic Genome — Multiple linear DNA chromosomes housed inside the membrane-bound nucleus .

Chromatin vs. Chromosome — Same DNA + histone complex, just different packing/condensation states !

Nucleotide Monomer & Backbone — Consists of a 5′ phosphate group , a 5-carbon pentose sugar (2′-deoxyribose in DNA; ribose with 2′-OH in RNA), and a 1′ nitrogenous base . Monomers link via...

Prokaryotic Genome — Single circular double-stranded DNA chromosome located in the cytoplasmic nucleoid region (no nuclear envelope; no histone nucleosomes in bacteria).

Plasmids (Extrachromosomal DNA) — Small, circular double-stranded DNA molecules found in bacteria (and rarely yeast/eukaryotes) that replicate independently, carry non-essential accessory genes...

Plasmids ≠ Organelles — Plasmids are small naked circular DNA rings, not membrane-bound organelles.

Antiparallel Double Helix — Two DNA strands run in opposite directions (one 5′ → 3′ , the other 3′ → 5′) with hydrophilic sugar-phosphate backbones outside and hydrophobic bases stacked...

§2. GENETIC MATERIAL ORGANIZATION: PROKARYOTES VS. EUKARYOTES

Chromosomes & Plasmids

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Genetic Material Organization ➔ **Universal Genetic Storage** ➔ **Prokaryotic Genome** ➔ **Eukaryotic Genome** ➔ **Plasmids (Extrachromosomal DNA)**

- **Universal Genetic Storage:** DNA is the primary hereditary material in **all cellular organisms** (prokaryotes & eukaryotes); some viruses use **RNA genomes** (e.g., retroviruses like HIV, influenza, SARS-CoV-2).
- **Prokaryotic Genome:**
  - Single **circular double-stranded DNA chromosome** located in the cytoplasmic **nucleoid region** (no nuclear envelope; no histone nucleosomes in bacteria).
- **Eukaryotic Genome:**
  - Multiple **linear DNA chromosomes** housed inside the membrane-bound **nucleus**.
  - Negatively charged DNA (**$-PO_4^{2-}$**) wraps tightly around positively charged **histone proteins** (rich in basic Lysine/Arginine) to form **nucleosomes** ("beads on a string") → coils into **chromatin** → condenses into visible **chromosomes** during division.
- **Plasmids (Extrachromosomal DNA):** Small, circular double-stranded DNA molecules found in **bacteria** (and rarely yeast/eukaryotes) that replicate independently, carry non-essential accessory genes (e.g., **antibiotic resistance**, toxin/metabolic genes), and can be transferred via **conjugation** or lab transformation.

[CORE AXIOM / REMEMBER] **Prokaryotes** = Circular chromosome + Cytoplasmic nucleoid + Plasmids.
Eukaryotes = Linear chromosomes + Nucleus + Histone packaging (chromatin).

[COMMON AP EXAM FRQ TRAP]

- **Chromatin vs. Chromosome:** Same DNA + histone complex, just different *packing/condensation states*!
- **Plasmids ≠ Organelles:** Plasmids are small naked circular DNA rings, *not* membrane-bound organelles.

[AP EXAM FRQ SCORING STRATEGY] Whenever an AP question mentions rapid spread of **antibiotic resistance** between bacteria or inserting a human gene into bacteria, immediately think **Plasmids**!

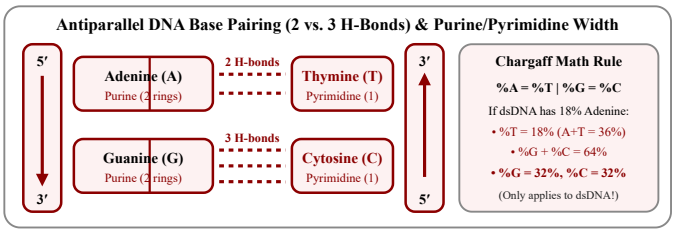
§4. DOUBLE HELIX POLARITY, PURINES/PYRIMIDINES & BASE PAIRING

5′→3′ & H-Bonds

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Double Helix Polarity, Purines/Pyrimidines & Base Pairing ➔
Nucleotide Monomer & Backbone ➔ **Antiparallel Double Helix**

- **Nucleotide Monomer & Backbone:** Consists of a **5′ phosphate group**, a **5-carbon pentose sugar** (2′-deoxyribose in DNA; ribose with 2′-OH in RNA), and a **1′ nitrogenous base**. Monomers link via covalent **phosphodiester bonds** (3′-OH to 5′-phosphate) forming the exterior backbone; the linear base sequence encodes genetic information.
- **Antiparallel Double Helix:** Two DNA strands run in opposite directions (one **5′ → 3′**, the other **3′ → 5′**) with hydrophilic sugar-phosphate backbones outside and hydrophobic bases stacked inside.
- **Purines (2 Rings) vs. Pyrimidines (1 Ring) & Chargaff's Rules:**
  - **Purines (Double-ring):** **Adenine (A)** and **Guanine (G)** — *"Pure As Gold (2 rings)"*.
  - **Pyrimidines (Single-ring):** **Cytosine (C)**, **Thymine (T, in DNA)**, **Uracil (U, in RNA)** — *"CUT the Py (1 ring)"*.
  - **Complementary Base Pairing (Uniform Helix Diameter):** A 2-ring purine always pairs with a 1-ring pyrimidine: **A = T (or A = U in RNA) via 2 H-bonds**; **G ≡ C via 3 H-bonds** (more G–C = higher thermal stability). Thus, **%A = %T** and **%G = %C**!



§3. CLASSIC EXPERIMENTS IDENTIFYING DNA AS HEREDITARY MATERIAL

Experimental Design

SCIENTIST(S)	EXPERIMENTAL SETUP & DISCOVERY
Griffith (1928)	Heat-killed virulent Smooth (S) *S. pneumoniae* mixed with live non-virulent Rough (R) bacteria transformed R cells into live, lethal S cells → discovered **bacterial transformation**.
Avery, MacLeod, McCarty (1944)	Treated heat-killed S extract with protease, RNase, or **DNase**. Only **DNase destroyed transforming ability** → proved **DNA** is the transforming principle.
Hershey & Chase (1952)	Bacteriophage T2 labeled with radioactive **^{32}P (DNA)** vs. **^{35}S (Protein coat)**. Only **^{32}P entered bacterial pellet** to direct new phage synthesis → confirmed DNA is viral genetic material.
Franklin, Watson & Crick (1953)	Rosalind Franklin's **X-ray diffraction (Photo 51)** revealed helical dimensions; Watson & Crick built the **antiparallel, complementary double-helix model**.

§5. DNA VS. RNA STRUCTURAL & FUNCTIONAL COMPARISON

Master Blueprint vs. Copy

FEATURE	DNA (DEOXYRIBONUCLEIC ACID)	RNA (RIBONUCLEIC ACID)
Pentose Sugar	**Deoxyribose** (lacks O at 2′ carbon: **2′-H**)	**Ribose** (has reactive hydroxyl at 2′ carbon: **2′-OH**)
Strandedness	**Double-stranded** (**dsDNA**) antiparallel helix	Usually **single-stranded** (**ssRNA**); folds into 3D shapes (tRNA/rRNA) via internal base pairing
Nitrogenous Bases	**A, G, C, Thymine (T)**	**A, G, C, Uracil (U)**
Chemical Stability	**More stable** (**2′-H** resists alkaline hydrolysis; double strand protects bases)	**Less stable / transient** (**2′-OH** makes RNA prone to catalytic self-cleavage)
Cellular Role	Long-term **"master blueprint"** storing hereditary info	Temporary **"working copy"**: **mRNA** (codons), **tRNA** (amino acid carrier), **rRNA** (catalytic ribosome)

[CORE AXIOM / REMEMBER] **Why H-Bonds Matter:** Hydrogen bonds between bases are **weak individually** (allowing helicase/RNA polymerase to unzip strands easily without breaking covalent phosphodiester bonds) but **strong collectively** across millions of base pairs!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.1 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 6.1: DNA & RNA Structure, Chromosomes, Plasmids & Classic Experiments ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (μg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.1 to prove why the claim holds.

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